

First-line Chemotherapy in Combination With Oral Recombinant Methioninase and a Low-methionine Diet for a Stage IV Inoperable Pancreatic-Cancer Patient Resulted in 40% Tumor Reduction and an 86% CA19-9 Biomarker Decrease

MOTOKAZU SATO^{1,2,3,4}, QINGHONG HAN¹, CHIHIRO HOZUMI⁵, HIDEO KUJIRAOKA⁵, KOHEI MIZUTA^{1,2}, SEI MORINAGA^{1,2}, BYUNG MO KANG^{1,2}, NORITOSHI KOBAYASHI³, YASUSHI ICHIKAWA³, ATSUSHI NAKAJIMA⁴ and ROBERT M. HOFFMAN^{1,2}

¹AntiCancer Inc., San Diego, CA, U.S.A.;

²Department of Surgery, University of California, San Diego, CA, U.S.A.;

³Department of Oncology, Yokohama City University School of Medicine, Yokohama, Japan;

⁴Department of Gastroenterology and Hepatology,

Yokohama City University School of Medicine, Yokohama, Japan;

⁵AntiCancer Japan Inc., Chiba, Japan

Abstract. *Background/Aim:* Pancreatic cancer has a very poor prognosis with a 5-year survival rate of less than 5% among patients with distant metastasis, a figure that has not improved over many decades. Only 10 to 20% patients are candidates for curative surgery at presentation due to the aggressive nature and asymptomatic progression of pancreatic cancer. Although first-line chemotherapy, such as FOLFIRINOX and gemcitabine + nab paclitaxel, improved the median survival from 8.5 to 11.1 months, more effective treatments are immediately needed. The aim of the present study was to evaluate the efficacy of methionine restriction with oral rMETase (o-rMETase) and a low-methionine diet combined with first-line chemotherapy on a patient with stage IV metastatic pancreatic cancer. *Case Report:* A 63-year-old female was diagnosed with metastatic pancreatic cancer in October 2023. The patient started FOLFIRINOX as first-line chemotherapy in combination with methionine restriction, which comprised o-rMETase 250 units twice a day and a low-methionine diet. The patient was monitored using

computed tomography and CA19-9 blood tests. After five months from the start of combination therapy, the size of the primary tumor decreased by 40% along with liver-metastasis regression. The CA19-9 blood marker decreased by 86%. The patient sustains a high performance status and continues the combination therapy without severe side effects. *Conclusion:* Methionine restriction consisting of o-rMETase and a low-methionine diet, in combination with first-line chemotherapy, was highly effective in a patient with inoperable stage IV pancreatic cancer.

The prognosis of pancreatic cancer is poor with a 5-year survival rate of approximately 12% (1) although many novel therapies have been introduced over five decades. Pancreatic cancer frequently progresses asymptotically, and only 10 to 20% of patients are eligible for surgery due to tumor invasion of adjacent organs or distant metastases at the time of presentation (1-3). Therefore, chemotherapy plays a significant role in pancreatic cancer treatment. Currently, FOLFIRINOX (5-fluorouracil/leucovorin, irinotecan, and oxaliplatin) and gemcitabine (GEM) + nab-paclitaxel (nab-PTX) are first-line chemotherapies for advanced pancreatic cancer. However, the median survival increased marginally from 8.5 to 11.1 months with these regimens (4, 5). More effective treatments are urgently needed.

Methionine addiction is a fundamental and general hallmark of cancer, termed as the Hoffman effect (6-9). Methionine addiction is at least partly due to excess transmethylation in cancer cells, leading to a significantly greater demand for methionine from external sources compared to normal cells, regardless of high-endogenous synthesis of methionine from homocysteine (6, 10-14). As a result, cancer cells are dependent

Correspondence to: Robert M. Hoffman, Ph.D., AntiCancer Inc., 7917 Ostrow St, Suite B, San Diego, CA 92111, U.S.A. Tel: +1 6198852284, e-mail: all@anticancer.com

Key Words: Pancreatic cancer, stage IV, methionine addiction, Hoffman effect, methionine restriction, oral methioninase, FOLFIRINOX, gemcitabine, nab paclitaxel, combination therapy, response, Hoffman protocol, synergy, CA19-9.



This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY-NC-ND) 4.0 international license (<https://creativecommons.org/licenses/by-nc-nd/4.0>).



Figure 1. Computed tomography images of pancreatic cancer before (10/20/23) and after (1/20/24 and 3/20/24) o-rMETase and a low-methionine diet were started in combination with first-line chemotherapy. Red circles denote the tumor circumference.

on exogenous methionine for survival, despite their high production of methionine (6, 12-14).

Our laboratory has developed recombinant methioninase (rMETase) to target methionine addiction (15). When administered orally, this enzyme breaks down methionine in the gut, resulting in a reduction in methionine levels in the blood and in tumors (16, 17). rMETase has shown efficacy in inhibiting pancreatic cancer in patient-derived orthotopic xenograft (PDOX) models (18).

Methionine restriction, including oral rMETase (o-rMETase) selectively arrests the cell cycle of cancer cells in the late-S/G2 phase (19, 20). Since S-phase is the main target of cytotoxic chemotherapy, it is synergistic with methionine restriction. There have been numerous reports that o-rMETase has synergistic efficacy with many chemotherapy drugs without side effects (21).

We previously showed that the biomarker level and tumor size of a stage IV inoperable pancreatic cancer patient treated with FOLFIRINOX in combination with o-rMETase and a low-methionine diet remained stable for 19 months with a high performance status (22).

In the present case report, a 63-year-old female with inoperable pancreatic cancer treated with first-line chemotherapy, combined with o-rMETase and a low-methionine diet, demonstrated a 40% tumor reduction, liver-metastasis regression, and an 86% of decrease in the pancreatic-cancer biomarker CA19-9 over five months.

Materials and Methods

Production and formulation of rMETase. rMETase was produced by fermenting recombinant *Escherichia coli* transformed with the *methioninase* gene from *Pseudomonas putida*. The purification of methioninase involved a high-yield method that had a heat step at 60 degrees Celsius, polyethylene glycol precipitation, and ion-exchange column chromatography using diethylaminoethyl (DEAE)-Sepharose FF (23,24).

Methionine restriction and rMETase administration. The patient went on a methionine-restricted diet, following the Nutritional Oncology Research Institute (NORI) protocol (25). This procedure recommends less than 2 mg/kg methionine intake per day. rMETase was administered orally twice a day approximately 30 min after each meal at a dose of 250 units as a supplement.

Case Report

A 63-year-old female was diagnosed with liver-metastatic pancreatic cancer in October 2023. The computed tomography (CT) images of October 2023 revealed that the patient had pancreatic cancer with the primary tumor highly invasive to adjacent organs and vasculature, and massive liver metastasis. The patient started FOLFIRINOX as first-line chemotherapy along with methionine restriction consisting of o-rMETase and a low-methionine diet. After the start of the combination therapy, the size of the primary tumor decreased by 40% and the liver metastasis shrunk over five months (Figure 1 and Figure 2). CA19-9, which is an important pancreatic cancer biomarker, also decreased by 86% (Figure 3). The patient experienced a repetitive neutropenia due to the side effects of FOLFIRINOX. Therefore, the regimen of chemotherapy was changed to GEM plus nab-PTX. The patient maintains a high-performance status and continues chemotherapy.

Discussion

A 40% decrease in the tumor size of a stage IV pancreatic cancer patient treated with standard chemotherapy is rare (5). Pancreatic cancer, especially cases with distant metastasis, have a poor prognosis, with reported 1-year survival rates of less than 10% (26), although the overall response rate is approximately 30% (5, 26). In the present case the patient

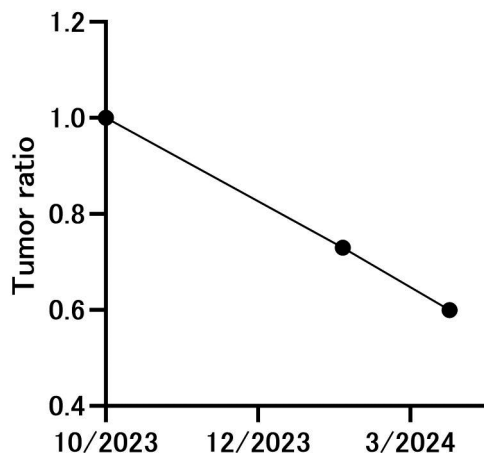


Figure 2. Ratio of tumor size with respect to size at the start of treatment of a pancreatic cancer patient with o-rMETase and a low-methionine diet along with first-line chemotherapy. A 40% reduction of the primary tumor was observed.

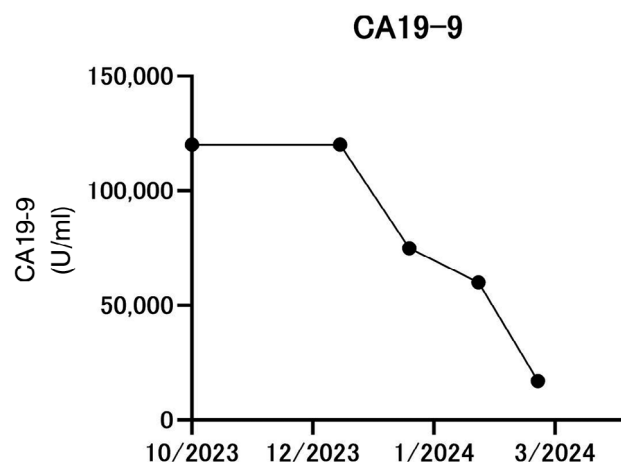


Figure 3. CA19-9, biomarker of pancreatic cancer patient. CA19-9 decreased by 86% after the start of o-rMETase and a low-methionine diet along with first-line chemotherapy.

achieved a 40% decrease in the primary tumor, a significant partial response (PR). The present patient has continued first-line treatment with GEM plus nab-PTX after FOLFIRINOX for seven months. Further observation is needed.

We first found that methionine restriction is synergistic with chemotherapy, termed the Hoffman protocol, almost 40 years ago (21, 27). There is synergy between methionine restriction, with rMETase and/or a low-methionine diet, and all types of chemotherapy (21). Cancer cells selectively arrest in the late-S/G₂ phase of the cell cycle in response to methionine depletion (19, 20). Hence, chemotherapy drugs targeting

S-phase exhibit enhanced efficacy when combined with methionine restriction.

o-rMETase is showing promise in the clinic for pancreatic cancer as shown in the present and previous study (22), and for others major cancers (28-36). Further clinical studies are necessary to test a combination of methionine restriction and first-line chemotherapy for stage IV pancreatic cancer.

Methionine addiction is a promising target for cancer therapy since it is tightly linked to malignancy (37-43) and is a universal hallmark of cancer (44-46).

Conflicts of Interest

The Authors declare no competing interests regarding this work.

Authors' Contributions

MS and RMH wrote the article. QH provided methioninase. CH and HK provided methioninase to the patient. MS, QH, KM, SM, BMK, NK, YI, AN, and RMH critically reviewed the article.

Acknowledgements

This paper is dedicated to the memory of A. R. Moossa, MD, Sun Lee, MD, Professor Gordon H. Sato, Professor Li Jiaxi, Masaki Kitajima, MD, Shigeo Yagi, PhD, Jack Geller, MD, Joseph R Bertino, MD, J.A.R. Mead, PhD, Eugene P. Frenkel, MD, Professor Sheldon Penman, Professor Lev Bergelson, Professor John R. Raper and Joseph Leighton, MD.

Funding

The Robert M. Hoffman Foundation for Cancer Research provided funds for this study.

References

- 1 Park W, Chawla A, O'Reilly EM: Pancreatic cancer: a review. *JAMA* 326(9): 851-862, 2021. DOI: 10.1001/jama.2021.13027
- 2 Ilic M, Ilic I: Epidemiology of pancreatic cancer. *World J Gastroenterol* 22(44): 9694-9705, 2016. DOI: 10.3748/wjg.v22.i44.9694
- 3 Vincent A, Herman J, Schulick R, Hruban RH, Goggins M: Pancreatic cancer. *Lancet* 378(9791): 607-620, 2011. DOI: 10.1016/S0140-6736(10)62307-0
- 4 Von Hoff DD, Ervin T, Arena FP, Chiorean EG, Infante J, Moore M, Seay T, Tjulandin SA, Ma WW, Saleh MN, Harris M, Reni M, Dowden S, Laheru D, Bahary N, Ramanathan RK, Tabernero J, Hidalgo M, Goldstein D, Van Cutsem E, Wei X, Iglesias J, Renschler MF: Increased survival in pancreatic cancer with nab-paclitaxel plus gemcitabine. *N Engl J Med* 369(18): 1691-1703, 2013. DOI: 10.1056/NEJMoa1304369
- 5 Conroy T, Desseigne F, Ychou M, Bouché O, Guimbaud R, Bécouarn Y, Adenis A, Raoul JL, Gourgou-Bourgade S, de la Fouchardière C, Bannoun J, Bachet JB, Khemissa-Akouf F, Péré-Vergé D, Delbaldo C, Assenat E, Chauffert B, Michel P, Montoto-Grillot C, Ducreux M, Groupe Tumeurs Digestives of Unicancer,

- PRODIGE Intergroup: FOLFIRINOX *versus* gemcitabine for metastatic pancreatic cancer. *N Engl J Med* 364(19): 1817-1825, 2011. DOI: 10.1056/NEJMoa1011923
- 6 Hoffman RM, Erbe RW: High *in vivo* rates of methionine biosynthesis in transformed human and malignant rat cells auxotrophic for methionine. *Proc Natl Acad Sci USA* 73(5): 1523-1527, 1976. DOI: 10.1073/pnas.73.5.1523
- 7 Coalson DW, Mecham JO, Stern PH, Hoffman RM: Reduced availability of endogenously synthesized methionine for S-adenosylmethionine formation in methionine-dependent cancer cells. *Proc Natl Acad Sci USA* 79(14): 4248-4251, 1982. DOI: 10.1073/pnas.79.14.4248
- 8 Stern PH, Mecham JO, Wallace CD, Hoffman RM: Reduced free-methionine in methionine-dependent SV40-transformed human fibroblasts synthesizing apparently normal amounts of methionine. *J Cell Physiol* 117(1): 9-14, 1983. DOI: 10.1002/jcp.1041170103
- 9 Kaiser P: Methionine dependence of cancer. *Biomolecules* 10(4): 568, 2020. DOI: 10.3390/biom10040568
- 10 Yamamoto J, Han Q, Inubushi S, Sugisawa N, Hamada K, Nishino H, Miyake K, Kumamoto T, Matsuyama R, Bouvet M, Endo I, Hoffman RM: Histone methylation status of H3K4me3 and H3K9me3 under methionine restriction is unstable in methionine-addicted cancer cells, but stable in normal cells. *Biochem Biophys Res Commun* 533: 1034-1038, 2020. DOI: 10.1016/j.bbrc.2020.09.108
- 11 Stern PH, Hoffman RM: Elevated overall rates of transmethylation in cell lines from diverse human tumors. *In Vitro* 20(8): 663-670, 1984. DOI: 10.1007/BF02619617
- 12 Wang Z, Yip LY, Lee JHJ, Wu Z, Chew HY, Chong PKW, Teo CC, Ang HY, Peh KLE, Yuan J, Ma S, Choo LSK, Basri N, Jiang X, Yu Q, Hillmer AM, Lim WT, Lim TKH, Takano A, Tan EH, Tan DSW, Ho YS, Lim B, Tam WL: Methionine is a metabolic dependency of tumor-initiating cells. *Nat Med* 25(5): 825-837, 2019. DOI: 10.1038/s41591-019-0423-5
- 13 Sullivan MR, Darnell AM, Reilly MF, Kunchok T, Joesch-Cohen L, Rosenberg D, Ali A, Rees MG, Roth JA, Lewis CA, Vander Heiden MG: Methionine synthase is essential for cancer cell proliferation in physiological folate environments. *Nat Metab* 3(11): 1500-1511, 2021. DOI: 10.1038/s42255-021-00486-5
- 14 Ghergurovich JM, Xu X, Wang JZ, Yang L, Ryseck RP, Wang L, Rabinowitz JD: Methionine synthase supports tumour tetrahydrofolate pools. *Nat Metab* 3(11): 1512-1520, 2021. DOI: 10.1038/s42255-021-00465-w
- 15 Hoffman RM: Development of recombinant methioninase to target the general cancer-specific metabolic defect of methionine dependence: a 40-year odyssey. *Expert Opin Biol Ther* 15(1): 21-31, 2015. DOI: 10.1517/14712598.2015.963050
- 16 Kawaguchi K, Miyake K, Han Q, Li S, Tan Y, Igarashi K, Kiyuna T, Miyake M, Higuchi T, Oshiro H, Zhang Z, Razmjooei S, Wangsiricharoen S, Bouvet M, Singh SR, Unno M, Hoffman RM: Oral recombinant methioninase (o-rMETase) is superior to injectable rMETase and overcomes acquired gemcitabine resistance in pancreatic cancer. *Cancer Lett* 432: 251-259, 2018. DOI: 10.1016/j.canlet.2018.06.016
- 17 Kawaguchi K, Han Q, Li S, Tan Y, Igarashi K, Miyake K, Kiyuna T, Miyake M, Chmielwski B, Nelson SD, Russell TA, Dry SM, Li Y, Singh AS, Eckardt MA, Unno M, Eilber FC, Hoffman RM: Intra-tumor L-methionine level highly correlates with tumor size in both pancreatic cancer and melanoma patient-derived orthotopic xenograft (PDOX) nude-mouse models. *Oncotarget* 9(13): 11119-11125, 2018. DOI: 10.18632/oncotarget.24264
- 18 Kawaguchi K, Miyake K, Han Q, Li S, Tan Y, Igarashi K, Lwin TM, Higuchi T, Kiyuna T, Miyake M, Oshiro H, Bouvet M, Unno M, Hoffman RM: Targeting altered cancer methionine metabolism with recombinant methioninase (rMETase) overcomes partial gemcitabine-resistance and regresses a patient-derived orthotopic xenograft (PDOX) nude mouse model of pancreatic cancer. *Cell Cycle* 17(7): 868-873, 2018. DOI: 10.1080/15384101.2018.1445907
- 19 Hoffman RM, Jacobsen SJ: Reversible growth arrest in simian virus 40-transformed human fibroblasts. *Proc Natl Acad Sci USA* 77(12): 7306-7310, 1980. DOI: 10.1073/pnas.77.12.7306
- 20 Yano S, Li S, Han Q, Tan Y, Bouvet M, Fujiwara T, Hoffman RM: Selective methioninase-induced trap of cancer cells in S/G2 phase visualized by Fucci imaging confers chemosensitivity. *Oncotarget* 5(18): 8729-8736, 2014. DOI: 10.18632/oncotarget.2369
- 21 Kubota Y, Han Q, Aoki Y, Masaki N, Obara K, Hamada K, Hozumi C, Wong ACW, Bouvet M, Tsunoda T, Hoffman RM: Synergy of combining methionine restriction and chemotherapy: the disruptive next generation of cancer treatment. *Cancer Diagn Progn* 3(3): 272-281, 2023. DOI: 10.21873/cdp.10212
- 22 Kubota Y, Han Q, Hozumi C, Masaki N, Yamamoto J, Aoki Y, Tsunoda T, Hoffman RM: Stage IV pancreatic cancer patient treated with FOLFIRINOX combined with oral methioninase: a highly-rare case with long-term stable disease. *Anticancer Res* 42(5): 2567-2572, 2022. DOI: 10.21873/anticancer.15734
- 23 Tan Y, Xu M, Tan X, Tan X, Wang X, Saikawa Y, Nagahama T, Sun X, Lenz M, Hoffman RM: Overexpression and large-scale production of recombinant L-methionine- α -deamino- γ -mercaptomethane-lyase for novel anticancer therapy. *Protein Expr Purif* 9(2): 233-245, 1997. DOI: 10.1006/prep.1996.0700
- 24 Takakura T, Ito T, Yagi S, Notsu Y, Itakura T, Nakamura T, Inagaki K, Esaki N, Hoffman RM, Takimoto A: High-level expression and bulk crystallization of recombinant L-methionine γ -lyase, an anticancer agent. *Appl Microbiol Biotechnol* 70(2): 183-192, 2006. DOI: 10.1007/s00253-005-0038-2
- 25 How to starve cancer naturally. Available at: <https://howtostarvecancer.naturally.com/methionine-chart> [Last accessed on December 22, 2023]
- 26 Survival for pancreatic cancer. Available at: <https://www.cancerresearchuk.org/about-cancer/pancreatic-cancer/survival> [Last accessed on May 15, 2024].
- 27 Stern PH, Hoffman RM: Enhanced *in vitro* selective toxicity of chemotherapeutic agents for human cancer cells based on a metabolic defect. *J Natl Cancer Inst* 76(4): 629-639, 1986. DOI: 10.1093/jnci/76.4.629
- 28 Sato M, Han Q, Mizuta K, Mori R, Kang BM, Morinaga S, Kobayashi N, Ichikawa Y, Nakajima A, Hoffman RM: Extensive Shrinkage and Long-term Stable Disease in a Teenage Female Patient With High-grade Glioma Treated With Temozolomide and Radiation in Combination With Oral Recombinant Methioninase and a Low-methionine Diet. *In Vivo* 38(3): 1459-1464, 2024. DOI: 10.21873/in vivo.13591
- 29 Kubota Y, Han Q, Hamada K, Aoki Y, Masaki N, Obara K, Tsunoda T, Hoffman RM: Long-term Stable Disease in a Rectal-cancer Patient Treated by Methionine Restriction With Oral Recombinant Methioninase and a Low-methionine Diet. *Anticancer Res* 42(8): 3857-3861, 2022. DOI: 10.21873/anticancer.15877
- 30 Han Q, Hoffman RM: Lowering and Stabilizing PSA Levels in Advanced-prostate Cancer Patients With Oral Methioninase. *Anticancer Res* 41(4): 1921-1926, 2021. DOI: 10.21873/anticancer.14958

- 31 Abo Qoura L, Balakin KV, Hoffman RM, Pokrovsky VS: The potential of methioninase for cancer treatment. *Biochim Biophys Acta Rev Cancer* 1879(4): 189122, 2024. DOI: 10.1016/j.bbcan.2024.189122
- 32 Sato M, Han Q, Mori R, Mizuta K, Kang BM, Morinaga S, Kobayashi N, Ichikawa Y, Nakajima A, Hoffman RM: Reduction of Tumor Biomarkers from very High to Normal and Extensive Metastatic Lesions to Undetectability in a Patient With Stage IV HER2-positive Breast Cancer Treated With Low-dose Trastuzumab Deruxtecan in Combination With Oral Recombinant Methioninase and a Low-methionine Diet. *Anticancer Res* 44(4): 1499-1504, 2024. DOI: 10.21873/anticancer.16946
- 33 Kubota Y, Han Q, Morinaga S, Tsunoda T, Hoffman RM: Rapid reduction of CEA and stable metastasis in an NRAS-mutant rectal-cancer patient treated with FOLFIRI and bevacizumab combined with oral recombinant methioninase and a low-methionine diet upon metastatic recurrence after FOLFIRI and bevacizumab treatment alone. *In Vivo* 37(5): 2134-2138, 2023. DOI: 10.21873/invivo.13310
- 34 Kubota Y, Han Q, Masaki N, Hozumi C, Hamada K, Aoki Y, Obara K, Tsunoda T, Hoffman RM: Elimination of axillary lymph-node metastases in a patient with invasive lobular breast cancer treated by first-line neo-adjuvant chemotherapy combined with methionine restriction. *Anticancer Res* 42(12): 5819-5823, 2022. DOI: 10.21873/anticancer.16089
- 35 Han Q, Hoffman RM: Chronic Treatment of an Advanced Prostate-cancer Patient With Oral Methioninase Resulted in Long-term Stabilization of Rapidly Rising PSA Levels. *In Vivo* 35(4): 2171-2176, 2021. DOI: 10.21873/invivo.12488
- 36 Han Q, Tan Y, Hoffman RM: Oral dosing of Recombinant Methioninase Is Associated With a 70% Drop in PSA in a Patient With Bone-metastatic Prostate Cancer and 50% Reduction in Circulating Methionine in a High-stage Ovarian Cancer Patient. *Anticancer Res* 40(5): 2813-2819, 2020. DOI: 10.21873/anticancer.14254
- 37 Hoffman RM, Jacobsen SJ, Erbe RW: Reversion to methionine independence in simian virus 40-transformed human and malignant rat fibroblasts is associated with altered ploidy and altered properties of transformation. *Proc Natl Acad Sci* 76(3): 1313-1317, 1979. DOI: 10.1073/pnas.76.3.1313
- 38 Hoffman RM, Jacobsen SJ, Erbe RW: Reversion to methionine independence by malignant rat and SV40-transformed human fibroblasts. *Biochem Biophys Res Commun* 82(1): 228-234, 1978. DOI: 10.1016/0006-291x(78)90600-9
- 39 Yamamoto J, Inubushi S, Han Q, Tashiro Y, Sugisawa N, Hamada K, Aoki Y, Miyake K, Matsuyama R, Bouvet M, Clarke SG, Endo I, Hoffman RM: Linkage of methionine addiction, histone lysine hypermethylation, and malignancy. *iScience* 25(4): 104162, 2022. DOI: 10.1016/j.isci.2022.104162
- 40 Aoki Y, Han Q, Tome Y, Yamamoto J, Kubota Y, Masaki N, Obara K, Hamada K, Wang JD, Inubushi S, Bouvet M, Clarke SG, Nishida K, Hoffman RM: Reversion of methionine addiction of osteosarcoma cells to methionine independence results in loss of malignancy, modulation of the epithelial-mesenchymal phenotype and alteration of histone-H3 lysine-methylation. *Front Oncol* 12: 1009548, 2022. DOI: 10.3389/fonc.2022.1009548
- 41 Yamamoto J, Aoki Y, Inubushi S, Han Q, Hamada K, Tashiro Y, Miyake K, Matsuyama R, Bouvet M, Clarke SG, Endo I, Hoffman RM: Extent and instability of trimethylation of histone H3 lysine increases with degree of malignancy and methionine addiction. *Cancer Genomics Proteomics* 19(1): 12-18, 2022. DOI: 10.21873/cgp.20299
- 42 Yamamoto J, Aoki Y, Han Q, Sugisawa N, Sun YU, Hamada K, Nishino H, Inubushi S, Miyake K, Matsuyama R, Bouvet M, Endo I, Hoffman RM: Reversion from methionine addiction to methionine independence results in loss of tumorigenic potential of highly-malignant lung-cancer cells. *Anticancer Res* 41(2): 641-643, 2021. DOI: 10.21873/anticancer.14815
- 43 Hoffman RM: Altered methionine metabolism, DNA methylation and oncogene expression in carcinogenesis. A review and synthesis. *Biochim Biophys Acta* 738: 49-87, 1984. DOI: 10.1016/0304-419x(84)90019-2
- 44 Mechem JO, Rowitch D, Wallace C, Stern PH, Hoffman RM: The metabolic defect of methionine dependence occurs frequently in human tumor cell lines. *Biochem Biophys Res Commun* 117(2): 429-434, 1983. DOI: 10.1016/0006-291x(83)91218-4
- 45 Stern PH, Wallace CD, Hoffman RM: Altered methionine metabolism occurs in all members of a set of diverse human tumor cell lines. *J Cell Physiol* 119(1): 29-34, 1984. DOI: 10.1002/jcp.1041190106
- 46 Tan Y, Xu M, Hoffman RM: Broad selective efficacy of recombinant methioninase and polyethylene glycol-modified recombinant methioninase on cancer cells *In Vitro*. *Anticancer Res* 30(4): 1041-1046, 2010

Received June 9, 2024

Revised June 24, 2024

Accepted June 26, 2024